



**NORTH
SYDNEY
GIRLS HIGH
SCHOOL**

2020

**Preliminary
Mathematics
Assessment
Task 2**

Preliminary Mathematics Advanced

General Instructions

- Reading Time – 2 minutes
- Working Time – 55 minutes
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- In Questions 6 – 9, show relevant mathematical reasoning and/or calculations

Total marks – 50

Section I – 5 marks (pages 2 – 3)

- Attempt Questions 1 – 5
- Allow about 6 minutes for this section

Section II – 45 marks (pages 5 – 20)

- Attempt Questions 6 – 9
- Allow about 49 minutes for this section

NAME: _____

TEACHER: _____

Question	1 – 5	6	7	8	9	Total
Algebra and Arithmetic	/2	/11	/2			/15
Functions	/3	/2	/9		/2	/16
Trigonometry				/11	/8	/19
	/5	/13	/11	/11	/10	/50

Section I

5 marks

Attempt Questions 1-5

Allow about 6 minutes for this section

Use the multiple choice answer sheet for Questions 1-5.

1 What is the value of $\frac{5\sqrt{\pi}}{\cos 28^\circ}$, correct to 2 significant figures?

- A. 10.04
- B. 10.03
- C. 1.04×10^1
- D. 1.0×10^1

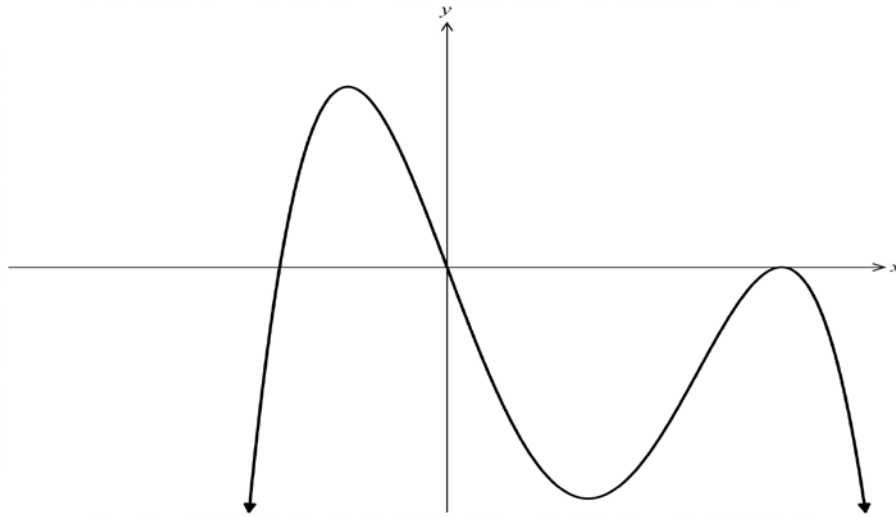
2 What is the simplest expression for $5^x \times 5^{2x}$?

- A. 5^{2x^2}
- B. 5^{3x}
- C. 25^{2x^2}
- D. 25^{3x}

3 What is the domain of $y = \sqrt{x-3} - \sqrt{7-x}$?

- A. $(-\infty, 3) \cup (7, \infty)$
- B. $(-\infty, 3] \cup [7, \infty)$
- C. $[3, 7]$
- D. $(3, 7)$

- 4 Consider the graph below.



Which of the following equations could represent the graph?

- A. $y = -3x(x-4)^2(x+2)$
- B. $y = 3x(x-4)^2(x+2)$
- C. $y = -3x(x+4)^2(x-2)$
- D. $y = 3x(x+4)^2(x-2)$
- 5 If $f(x-1) = x^2 - 2x + 3$, then which of the following is an expression for $f(x)$?
- A. $x^2 - 2$
- B. $x^2 + 2$
- C. $x^2 - 2x + 2$
- D. $x^2 - 2x + 4$

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Section II

50 marks

Attempt Questions 6 to 9.

Allow about 49 minutes for this section.

In questions 6 to 9, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (13 marks)

(a) Given that $\log_a b = 4.25$ and $\log_a c = 0.75$, find the value of:

(i) $\log_a \left(\frac{c}{b} \right)$ 1

(ii) $\log_a (bc)^3$. 1

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(b) Solve $|4 - 3x| = 7$ 2

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Question 6 continues on Page 6

Question 6 (continued)

- (c) Write $\frac{1+\sqrt{xy}}{x\sqrt{x}}$ in simplest form with a rational denominator, given $x, y > 0$ 2

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- (d) Solve $2x^2 - 2x - 1 = 0$. 2

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Question 6 continues on Page 7

Question 6 (continued)

(e) Solve the following pair of simultaneous equations:

3

$$7^{y-2x} = \frac{1}{7}$$

$$x^2 - 2x + y = 0$$

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Question 6 continues on Page 8

(f) The resistance to the motion of a train R is the sum of two parts, one is a constant of 300 Newtons and the other is directly proportional to square of its speed, v m/s. If the total resistance is 350 Newtons when the speed of the train is 30 m/s, calculate the total resistance when the speed is 36 m/s.

[illegible]

Question 7 (11 marks)

(a) Solve $12 + 5x - 2x^2 \leq 0$

2

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(b) The equation of a circle is $x^2 + y^2 - 2x + 4y - 20 = 0$. Find the circle's centre and radius.

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Question 7 continues on Page 10

(c) A function is defined as:

(i) Evaluate $f(-3) + f(2) - f(0)$. **2**

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- 10 -

Question 7 (continued)

(d) Let $f(x) = 3 + \sqrt{4 - x}$ and $g(x) = x^2$

- (i) find an expression for $f(g(x))$ **1**
- (ii) find the domain of $f(g(x))$ **1**
- (iii) sketch $y = f(g(x))$. **2**

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[illegible]

Question 8 (11 marks)

- (a) Express 495° in radians as an exact value. **1**

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- (b) Find the exact value of $\csc\left(-\frac{4\pi}{3}\right)$. **1**

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- (c) If $\cos\alpha = \frac{2}{5}$ and α is a reflex angle, find the exact value of $\tan\alpha$. **2**

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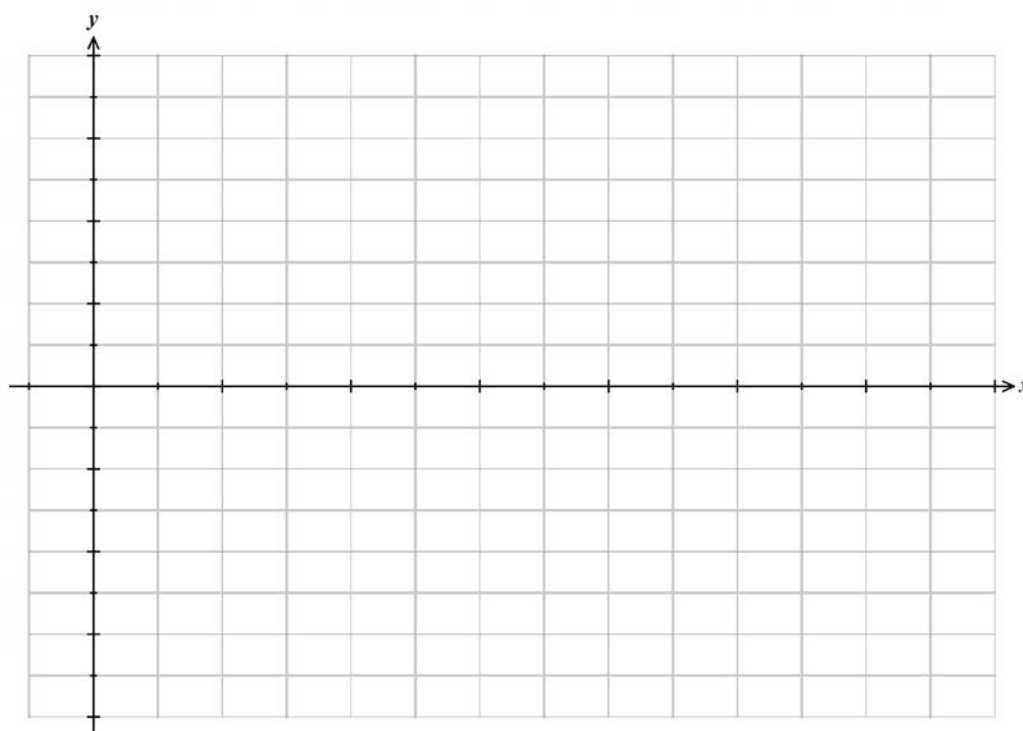
Question 8 continues on Page 14

Question 8 (continued)

(d) Show that $\frac{1}{\sin \alpha} - \frac{1}{\sin^2 \alpha} = \frac{\sin \alpha - 1}{(1 + \cos \alpha)(1 - \cos \alpha)}$. 2

[illegible]

(e) Sketch $y = 2 + \sin\left(\frac{\pi x}{3}\right)$ for $0 \leq x \leq 12$, showing all important features. 2



Question 8 continues on Page 15

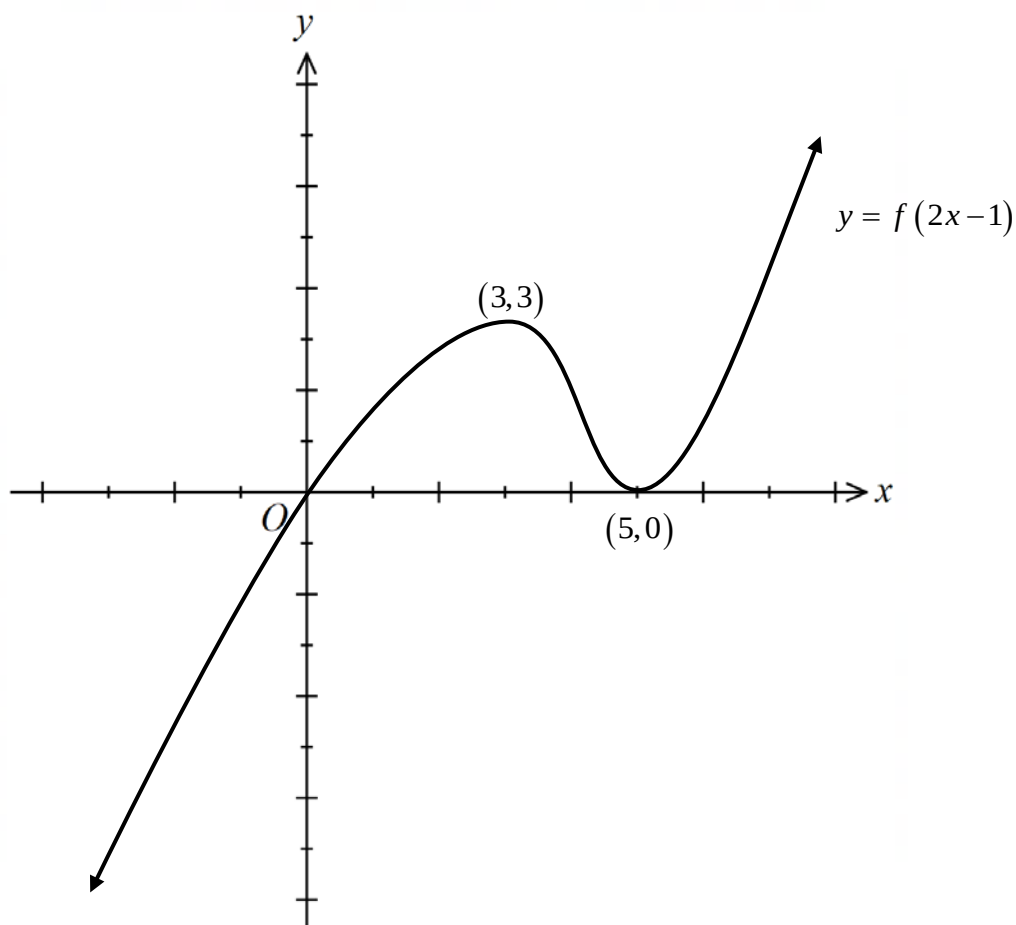
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Question 9 (10 marks)

- (a) The graph of the curve $y = f(2x - 1)$ is shown below.

2

The graph passes through the origin and points $(3, 3)$ and $(5, 0)$ respectively as shown.



By using the graph, or otherwise:

- (a) Find the number of solutions to the equation $f(2x - 1) - |x| = 0$.

1

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- (b) Solve $f(x) = 0$

1

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Question 9 continues on Page 18

(b) $\triangle ABC$ is an equilateral triangle with sides of length 6 m.

A diagram of a triangle ABC with vertex A at the top and base BC at the bottom. A dashed line segment connects vertex A to the base BC . The base BC is divided into two segments, BX and CY , which are shaded blue. A right-angle symbol is shown at the intersection of the dashed line and the base BC .

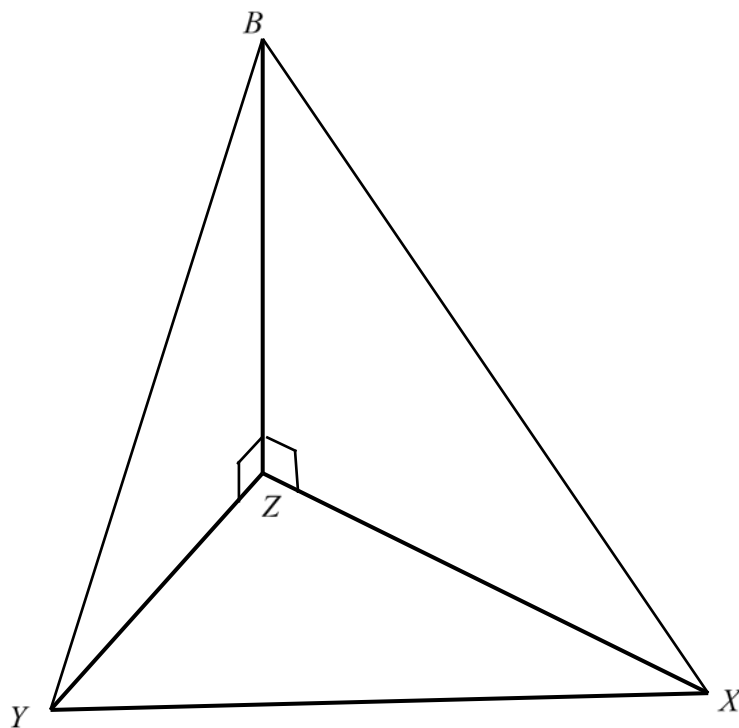
- (i) Show that the radius of the arc is $3\sqrt{3}$ m. **1**
- (ii) Find the exact area of the shaded region. **3**

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Question 9 continues on Page 19

Question 9 (continued)

- (c) The angle of elevation of a hot air balloon B from a point on the ground X is 29° and the bearing is 323° .
From point Y 4000 m due west of X , the bearing of the balloon is 049° .



- | | |
|--|---|
| (i) Draw $\triangle XYZ$ showing the bearings. | 1 |
| (ii) Explain why $\angle YZX = 86^\circ$. | 1 |
| (iii) Find the altitude of the balloon. | 2 |

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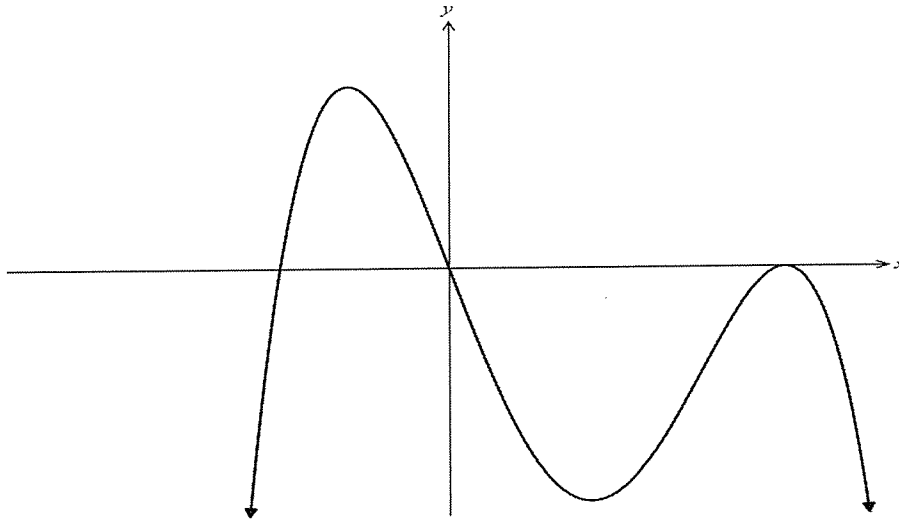
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Which of the following equations could represent the graph?

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- B. $y = 3x(x-4)^2(x+2)$
- C. $y = -3x(x+4)^2(x-2)$
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Question 6 (13 marks)

(a) Given that $\log_a b = 4.25$ and $\log_a c = 0.75$, find the value of:

(i) $\log_a \left(\frac{c}{b} \right)$

1

(ii) $\log_a (bc)^3$

1

(i) $\log_a c - \log_a b = 0.75 - 4.25$
 $= -3.5$

(ii) $3 \log_a bc = 3(\log_a b + \log_a c)$
 $= 3(4.25 + 0.75)$
 $= 15$

(b) Solve $|4 - 3x| = 7$

2

$4 - 3x = 7$ or $-4 + 3x = 7$
 $3x = -3$ $3x = 11$
 $x = -1$ $x = \frac{11}{3}$

Question 6 continues on Page 6

Question 6 (continued)

- (c) Write $\frac{1+\sqrt{xy}}{x\sqrt{x}}$ in simplest form with a rational denominator, given $x, y > 0$

2

$$\begin{aligned} & \frac{(1 + \sqrt{xy})}{x\sqrt{x}} \times \frac{\sqrt{x}}{\sqrt{x}} \\ &= \frac{\sqrt{x} + \sqrt{x} \cdot \sqrt{xy}}{x \cdot x} \\ &= \frac{\sqrt{x} + x\sqrt{y}}{x^2} \end{aligned}$$

- (d) Solve $2x^2 - 2x - 1 = 0$.

2

$$\begin{aligned} x &= \frac{2 \pm \sqrt{(-2)^2 - 4(2)(-1)}}{4} \\ &= \frac{2 \pm \sqrt{12}}{4} \\ &= \frac{2 \pm 2\sqrt{3}}{4} \\ &= \frac{1 \pm \sqrt{3}}{2} \end{aligned}$$

Question 6 continues on Page 7

Question 6 (continued)

(e) Solve the following pair of simultaneous equations:

3

$$7^{y-2x} = \frac{1}{7} = 7^{-1} \quad \text{--- ①}$$

$$x^2 - 2x + y = 0$$

From ① $\Rightarrow y - 2x = -1$

$$y = 2x - 1$$

From ② $\Rightarrow$

$$y = -x^2 + 2x$$

$$-x^2 + 2x = 2x - 1$$

$$x^2 = 1$$

$$x = 1 \text{ or } x = -1$$

$$x = -1$$

$$y = 2(1) - 1$$

$$y = 2(-1) - 1$$

$$y = 1$$

$$y = -3$$

$\therefore$ solutions are $x=1, y=1$ or $x=-1, y=-3$

Question 6 continues on Page 8

Question 6 (continued)

- (f) The resistance to the motion of a train R is the sum of two parts, one is a constant of 300 Newtons and the other is directly proportional to square of its speed, v m/s.

2

If the total resistance is 350 Newtons when the speed of the train is 30 m/s, calculate the total resistance when the speed is 36 m/s.

$$R = 300 + kV^2$$

find k : when $R = 350$, $V = 30$

$$350 = 300 + k \times 30^2$$

$$50 = 900k$$

$$k = \frac{50}{900} = \frac{1}{18}$$

$$\therefore R = 300 + \frac{V^2}{18}$$

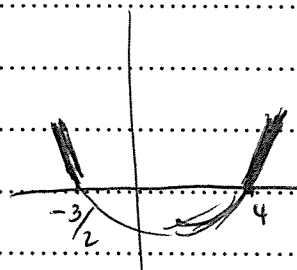
when $V = 36$, $R = 300 + \frac{36^2}{18}$

$$= 372 \text{ newtons}$$

Question 7 (11 marks)

(a) Solve $12 + 5x - 2x^2 \leq 0$

$$\begin{aligned} & \dots\dots\dots 2x^2 - 5x - 12 \geq 0 \dots\dots\dots \text{or} \dots\dots\dots (3 + 2x)(4 - x) \leq 0 \\ & \dots\dots\dots \begin{matrix} 2x^2 & & -5x & & -12 \\ 2x & & -4 & & 3 \end{matrix} \\ & \dots\dots\dots (x - 4)(2x + 3) \geq 0 \\ & \dots\dots\dots x \geq 4 \text{ or } x \leq -\frac{3}{2} \end{aligned}$$



(b) The equation of a circle is $x^2 + y^2 - 2x + 4y - 20 = 0$. Find the circle's centre and radius.

$$\begin{aligned} & \dots\dots\dots x^2 + 2x + 1 + y^2 + 4y + 4 = 20 + 1 + 4 \\ & \dots\dots\dots (x - 1)^2 + (y + 2)^2 = 25 \end{aligned}$$

$$\begin{aligned} & \dots\dots\dots \text{radius} = 5 \\ & \dots\dots\dots \text{centre} = (1, -2) \end{aligned}$$

Question 7 continues on Page 10

Question 7 (continued)

(c) A function is defined as:

$$f(x) = \begin{cases} 5x - 4 & \text{when } x \leq 0 \\ -x^2 & \text{when } x > 0 \end{cases}$$

(i) Evaluate $f(-3) + f(2) - f(0)$.

2

(ii) Write an expression for $f(a^2 + 2)$.

1

$$\begin{aligned} \text{(i)} \quad f(-3) &= 5(-3) - 4 = -19 \\ f(2) &= -(2)^2 = -4 \\ f(0) &= -4 \end{aligned}$$

$$\begin{aligned} f(-3) + f(2) - f(0) &= -19 - 4 + 4 \\ &= -19 \end{aligned}$$

$$\text{(ii)} \quad f(a^2 + 2) = -(a^2 + 2)^2 \text{ or } -a^4 - 4a^2 - 4$$

$$\text{as } a^2 + 2 \geq 0$$

Question 7 continues on Page 11

Question 7 (continued)

(d) Let $f(x) = 3 + \sqrt{4-x}$ and $g(x) = x^2$

(i) find an expression for $f(g(x))$

1

(ii) find the domain of $f(g(x))$

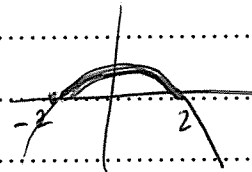
1

(iii) sketch $y = f(g(x))$.

2

(i) $f(g(x)) = f(x^2)$
 $= 3 + \sqrt{4 - x^2}$

(ii) $4 - x^2 \geq 0$
 $(2-x)(2+x) \geq 0$

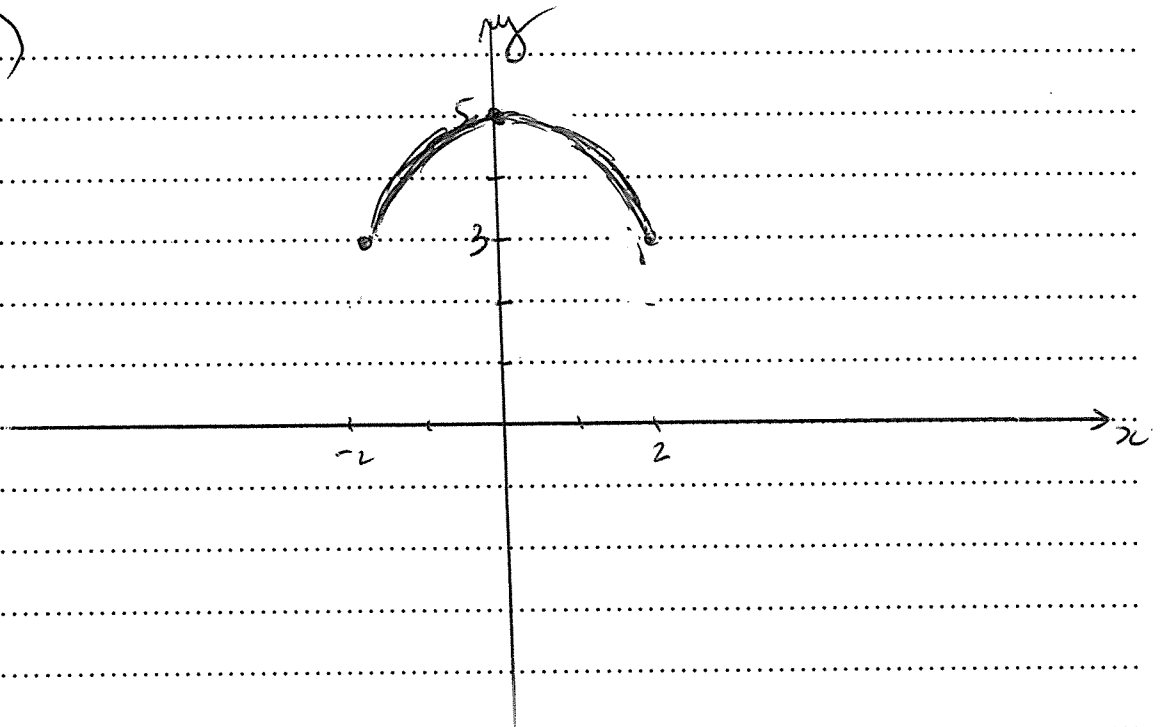


$-2 \leq x \leq 2$

and Domain of $g(x) = x^2 : -\infty < x < \infty$

$\therefore$ Domain of $f(g(x)) : -2 \leq x \leq 2$

(iii)



This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

Question 8 (11 marks)

- (a) Express 495° in radians as an exact value.

1

$$495^\circ \times \frac{\pi}{180} = \frac{11\pi}{4}$$

- (b) Find the exact value of $\csc\left(-\frac{4\pi}{3}\right)$.

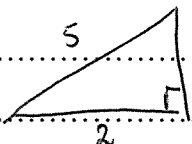
1

$$= \csc\left(\pi - \frac{\pi}{3}\right)$$

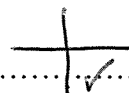
$$= \csc \frac{\pi}{3} = \frac{2}{\sqrt{3}}$$

- (c) If $\cos \alpha = \frac{2}{5}$ and α is a reflex angle, find the exact value of $\tan \alpha$.

2



$$\sqrt{25 - 4} = \sqrt{21}$$



$$\tan \alpha = -\frac{\sqrt{21}}{2}$$

Question 8 continues on Page 14

Question 8 (continued)

- (d) Show that $\frac{1}{\sin \alpha} - \frac{1}{\sin^2 \alpha} = \frac{\sin \alpha - 1}{(1 + \cos \alpha)(1 - \cos \alpha)}$. 2

$$\text{LHS} = \frac{1}{\sin \alpha} \times \frac{\sin \alpha}{\sin \alpha} = \frac{1}{\sin^2 \alpha}$$

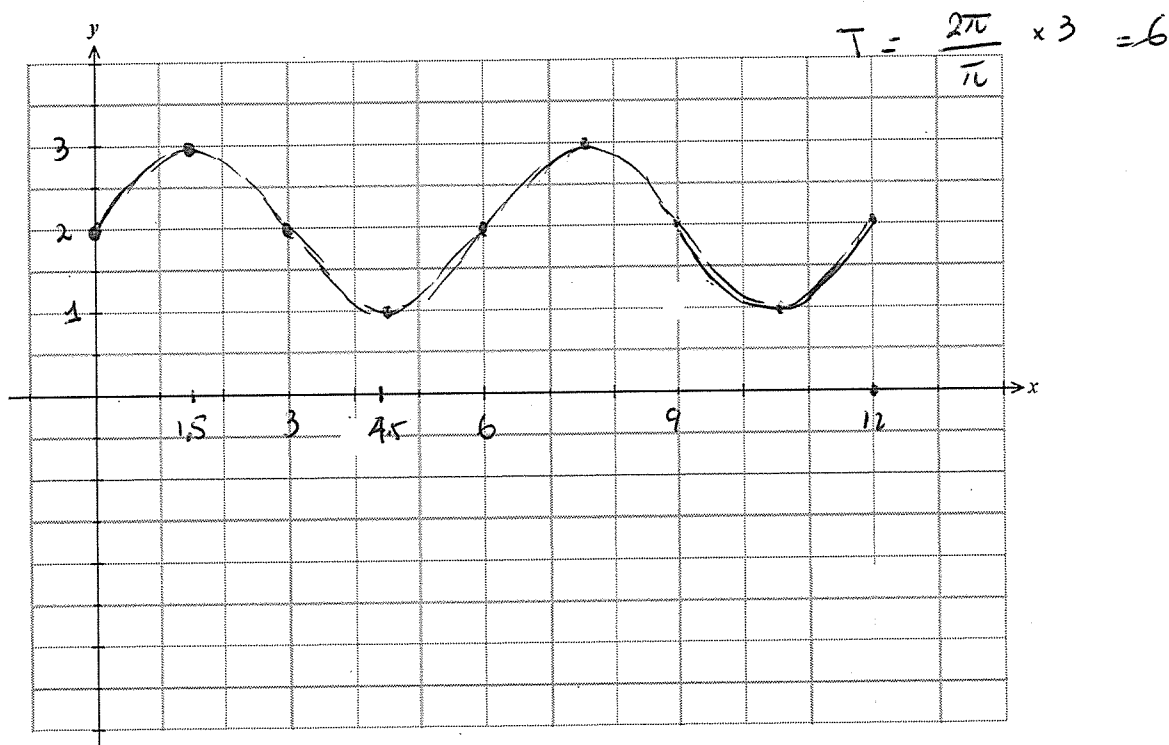
$$= \frac{\sin \alpha - 1}{\sin^2 \alpha}$$

$$= \frac{\sin \alpha - 1}{1 - \cos^2 \alpha}$$

$$= \frac{\sin \alpha - 1}{(1 - \cos \alpha)(1 + \cos \alpha)}$$

$$= \text{RHS}$$

- (e) Sketch $y = 2 + \sin\left(\frac{\pi x}{3}\right)$ for $0 \leq x \leq 12$, showing all important features. 2



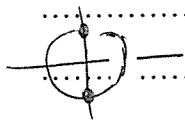
Question 8 continues on Page 15

Question 8 (continued)

- (f) Solve $4\sin^2 x + 7\cos x = 4$, for $0^\circ \leq x \leq 360^\circ$.
Give your answers correct to the nearest minute.

3

$$\begin{aligned}
 4\sin^2 x + 7\cos x - 4 &= 0 \\
 4(1 - \cos^2 x) + 7\cos x - 4 &= 0 \\
 4 - 4\cos^2 x + 7\cos x - 4 &= 0 \\
 -4\cos^2 x + 7\cos x &= 0 \\
 \cos x [7 - 4\cos x] &= 0
 \end{aligned}$$



$$\begin{aligned}
 \cos x &= 0 \\
 x &= 90^\circ, 270^\circ
 \end{aligned}$$

$$\begin{aligned}
 4\cos x &= 7 \\
 \cos x &= \frac{7}{4}
 \end{aligned}$$

↑

No solution

$$\text{as } -1 \leq \cos x \leq 1$$

[illegible]

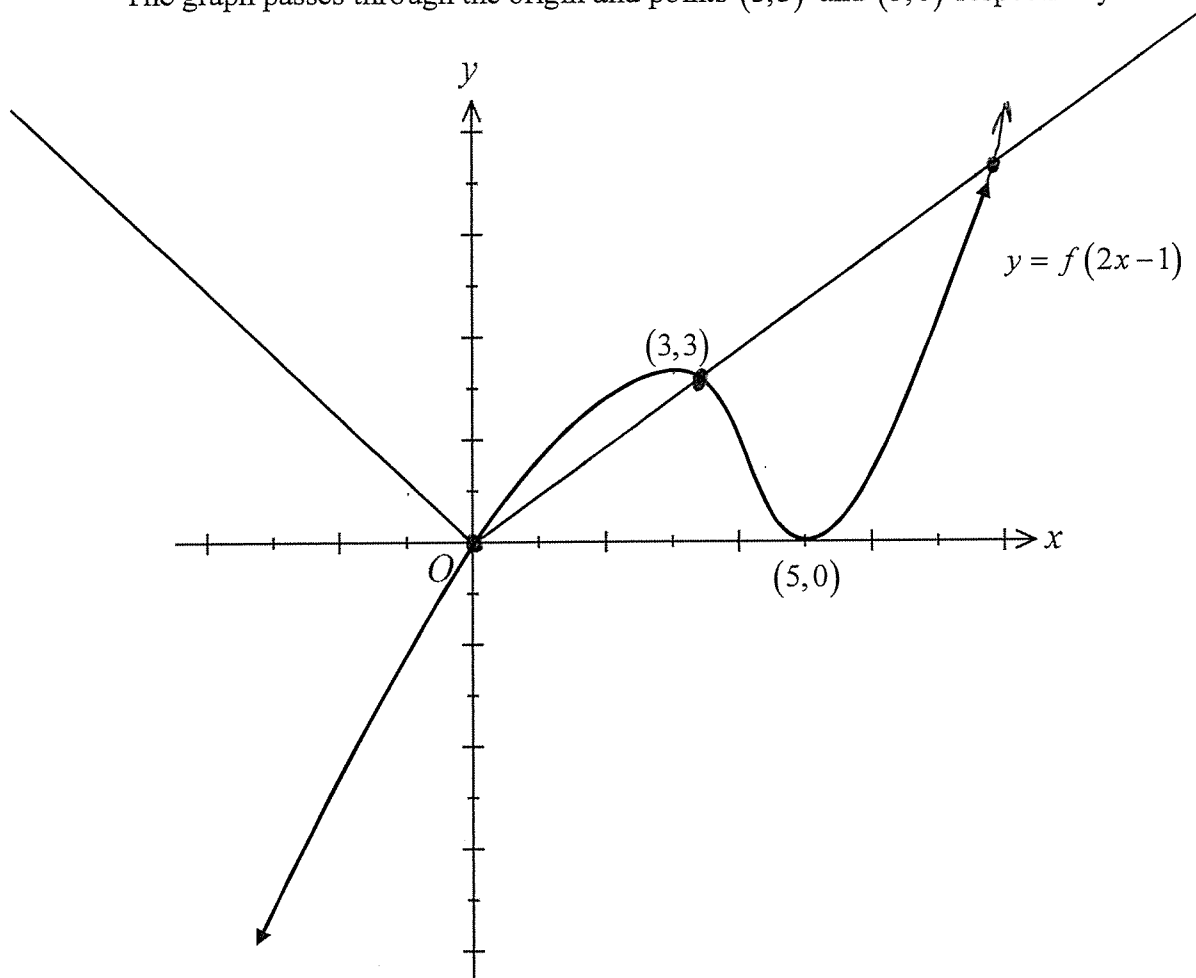
Question 9 (10 marks)

- (a) The graph of the curve $y = f(2x-1)$ is shown below.

$$= f\left(2\left(x-\frac{1}{2}\right)\right)$$

The graph passes through the origin and points $(3,3)$ and $(5,0)$ respectively as shown.

2



By using the graph, or otherwise:

- (a) Find the number of solutions to the equation $f(2x-1) - |x| = 0$.

1

$f(2x-1) = |x|$
three solutions

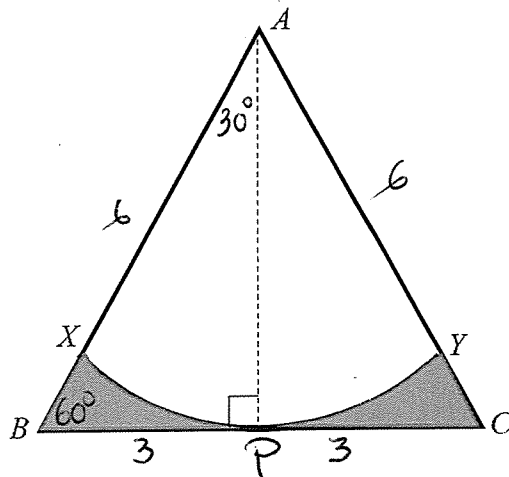
- (b) Solve $f(x) = 0$ two solutions $f(2x-1) = 0$ at $x = 1, 5$
 $f(2 \times 5 - 1) = f(9) = 0$
 $f(2 \times 0 - 1) = f(-1) = 0$
 $x = -1, 9$

Question 9 continues on Page 18

Question 9 (continued)

- (b) $\triangle ABC$ is an equilateral triangle with sides of length 6 m.

An arc with centre A and BC as tangent cuts AB and AC at X and Y respectively.



- (i) Show that the radius of the arc is $3\sqrt{3}$ m.

1

- (ii) Find the exact area of the shaded region.

3

$$\sin 60^\circ = \frac{AP}{6}$$

$$\frac{\sqrt{3}}{2} = \frac{AP}{6} \Rightarrow AP = \frac{6\sqrt{3}}{2} = 3\sqrt{3} \text{ m}$$

$$\text{ii) shaded A} = \text{Area } \triangle - \text{Area sector} \\ = \frac{6 \times 3\sqrt{3}}{2} - \frac{1}{2} \times \frac{\pi}{3} \times (3\sqrt{3})^2$$

$$= 9\sqrt{3} - \frac{1}{2} \times \frac{\pi}{3} \times 9 \times 3$$

$$= 9\sqrt{3} - \frac{9\pi}{2} \text{ or equivalent}$$

$$\text{OR shaded A} = \frac{6 \times 6 \times \sin \frac{\pi}{3}}{2} - \frac{1}{2} \times (3\sqrt{3})^2 \times \frac{\pi}{3}$$

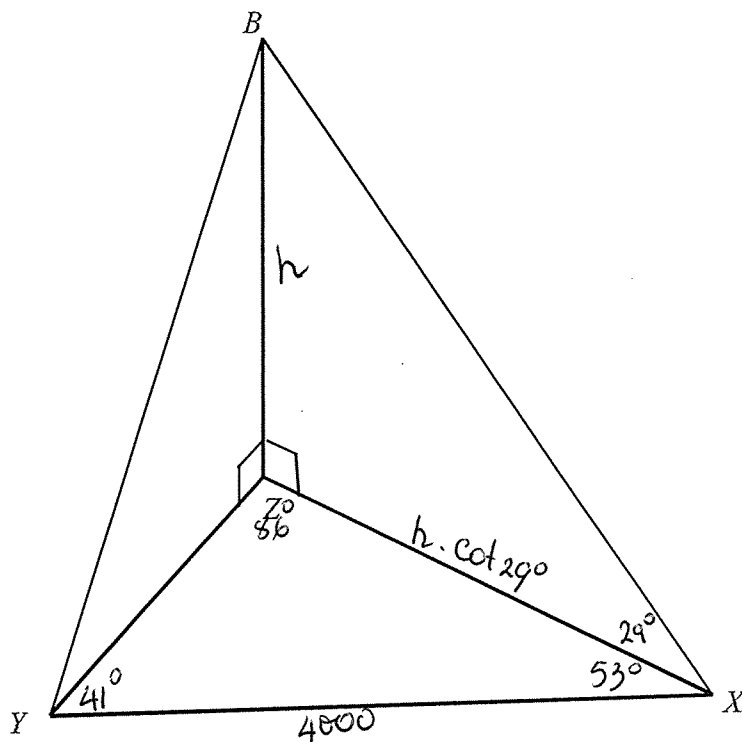
$$= 18 \times \frac{\sqrt{3}}{2} - \frac{9\pi}{2}$$

$$= 9\sqrt{3} - \frac{9\pi}{2}$$

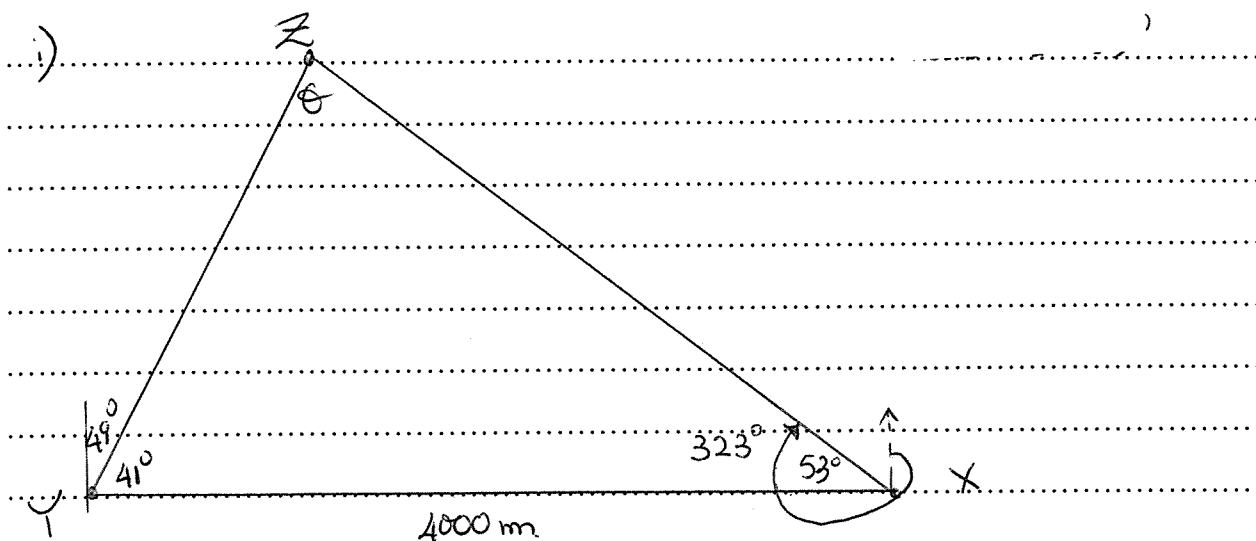
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Question 9 (continued)

- (c) The angle of elevation of a hot air balloon B from a point on the ground X is 29° and the bearing is 323° .
From point Y 4000 m due west of X , the bearing of the balloon is 049° .



- (i) Draw $\triangle XYZ$ showing the bearings. 1
(ii) Explain why $\angle YZX = 86^\circ$. 1
(iii) Find the altitude of the balloon. 2



- ii) let $\angle YZX = \theta$
 $\therefore \theta = 180^\circ - 53^\circ - 41^\circ = 86^\circ$

iii) let BZ be h

In ΔBXZ

$$\tan 29^\circ = \frac{h}{XZ}$$

$$XZ = \frac{h}{\tan 29^\circ} = h \cot 29^\circ$$

In ΔXYZ :

Using Sine Rule

$$\frac{h \cot 29^\circ}{\sin 41^\circ} = \frac{4000}{\sin 86^\circ}$$

$$h = \frac{4000 \times \sin 41^\circ}{\sin 86^\circ \times \cot 29^\circ}$$

$$h = \frac{4000 \times \sin 41^\circ \times \tan 29^\circ}{\sin 86^\circ}$$

$$= 1458.2 \text{ m}$$

$$\approx 1459 \text{ m}$$

OR

$$\frac{h / \tan 29^\circ}{\sin 41^\circ} = \frac{4000}{\sin 86^\circ}$$

$$\frac{h}{\tan 29^\circ \sin 41^\circ} = \frac{4000}{\sin 86^\circ}$$

$$h = \frac{4000 \cdot \sin 41^\circ \cdot \tan 29^\circ}{\sin 86^\circ}$$

$$= 1459 \text{ m}$$

End of Paper